📄 **Project: Simple Billing System**

🎯 **Goal:**

Make a small program that calculates a customer's total bill — including tax — using **user-defined functions**.

🧠 **Concept Used:**

- Function definition (def)
- Function calling
- Parameters and return values

💻 **Code Example:**

```
# Simple Billing System


# Function to calculate total amount

def calculate_total(price, quantity, tax_rate):

    subtotal = price * quantity

    tax = subtotal * tax_rate / 100

    total = subtotal + tax

    return total


# Function to display bill

def print_bill(item_name, price, quantity, tax_rate):

    total = calculate_total(price, quantity, tax_rate)

    print("------ BILL ------")

    print(f"Item: {item_name}")

    print(f"Price per item: ₹{price}")
```

```
    print(f"Quantity: {quantity}")
    print(f"Tax rate: {tax_rate}%")
    print("------------------")
    print(f"Total Amount: ₹{total:.2f}")
    print("------------------")

# Main program
item = input("Enter item name: ")
price = float(input("Enter price per item: ₹"))
quantity = int(input("Enter quantity: "))
tax = float(input("Enter tax rate (%): "))

print_bill(item, price, quantity, tax)
```

🧮 **Project: Student Grade Calculator**

🎯 **Goal:**

Calculate and display a student's grade based on their marks using a **function**.

💻 **Code Example:**

```
# Student Grade Calculator
```

```
# Function to calculate grade
def calculate_grade(marks):
    if marks >= 90:
        return "A+"
    elif marks >= 80:
        return "A"
    elif marks >= 70:
        return "B"
    elif marks >= 60:
        return "C"
    elif marks >= 50:
        return "D"
    else:
        return "Fail"

# Function to display result
def show_result(name, marks):
    grade = calculate_grade(marks)
    print("----- RESULT -----")
    print(f"Student Name: {name}")
    print(f"Marks: {marks}")
    print(f"Grade: {grade}")
    print("------------------")

# Main program
```

```
student_name = input("Enter student name: ")

student_marks = float(input("Enter total marks (out of 100): "))


show_result(student_name, student_marks)
```

🧩 **Example Output:**

```
Enter student name: Riya

Enter total marks (out of 100): 85


----- RESULT -----

Student Name: Riya

Marks: 85.0

Grade: A

-----------
```

🕒 **Project: Digital Clock (Using Function)**

🎯 **Goal:**

Show the current time updating every second using a **user-defined function**.

💻 **Code Example:**

```
import time

from datetime import datetime
```

```
# Function to display current time

def show_time():

    while True:

        current_time = datetime.now().strftime("%H:%M:%S")

        print("Current Time:", current_time, end="\r")  # overwrites same line

        time.sleep(1)  # updates every second


# Main program

print("Press Ctrl+C to stop the clock.\n")

show_time()
```

🏦 **Project: Bank Account Management System**

🎯 **Goal:**

Create a simple program to deposit, withdraw, and check balance using functions.

💻 **Code Example:**

```
# Bank Account Management System


balance = 0  # global variable to store balance


# Function to deposit money
```

```
def deposit(amount):
    global balance
    balance += amount
    print(f"₹{amount} deposited successfully!")

# Function to withdraw money
def withdraw(amount):
    global balance
    if amount <= balance:
        balance -= amount
        print(f"₹{amount} withdrawn successfully!")
    else:
        print("Insufficient balance!")

# Function to check balance
def check_balance():
    print(f"Current Balance: ₹{balance}")

# Main program
while True:
    print("\n------ BANK MENU ------")
    print("1. Deposit Money")
    print("2. Withdraw Money")
    print("3. Check Balance")
    print("4. Exit")
```

```
choice = input("Enter your choice (1-4): ")


if choice == "1":
    amt = float(input("Enter amount to deposit: ₹"))
    deposit(amt)
elif choice == "2":
    amt = float(input("Enter amount to withdraw: ₹"))
    withdraw(amt)
elif choice == "3":
    check_balance()
elif choice == "4":
    print("Thank you for using our bank system!")
    break
else:
    print("Invalid choice! Please try again.")
```

👇

🧹 **Project: To-Do List Manager**

🎯 **Goal:**

Create a small **task management system** where the user can add, view, and delete tasks — using **user-defined functions**.

💻 **Code Example:**

```
# Simple To-Do List Manager


# Global list to store tasks

tasks = []


# Function to add a task

def add_task(task):

    tasks.append(task)

    print(f"Task '{task}' added successfully!")


# Function to view all tasks

def view_tasks():

    if len(tasks) == 0:

        print("No tasks available.")

    else:

        print("\n------ YOUR TASKS ------")

        for i, task in enumerate(tasks, start=1):

            print(f"{i}. {task}")

        print("------------------------")


# Function to delete a task

def delete_task(task_number):
```

```
    if 0 < task_number <= len(tasks):
        removed = tasks.pop(task_number - 1)
        print(f"Task '{removed}' deleted successfully!")
    else:
        print("Invalid task number!")

# Main program
while True:
    print("\n----- TO-DO MENU -----")
    print("1. Add Task")
    print("2. View Tasks")
    print("3. Delete Task")
    print("4. Exit")

    choice = input("Enter your choice (1-4): ")

    if choice == "1":
        task = input("Enter new task: ")
        add_task(task)
    elif choice == "2":
        view_tasks()
    elif choice == "3":
        view_tasks()
        num = int(input("Enter task number to delete: "))
        delete_task(num)
```

```
    elif choice == "4":
        print("Thank you for using To-Do Manager!")
        break
    else:
        print("Invalid choice! Try again.")
```

🎲 **Project: Dice Roller Game**

🎯 **Goal:**

Simulate rolling a dice — the user can roll multiple times — using **functions**.

💻 **Code Example:**

```
import random
import time

# Function to roll the dice
def roll_dice():
    return random.randint(1, 6)

# Function to play the dice game
def play_game():
    print("🎲 Welcome to the Dice Roller Game! 🎲")
    while True:
        choice = input("\nPress 'r' to roll or 'q' to quit: ").lower()
```

```
    if choice == 'r':
        print("Rolling the dice...")
        time.sleep(1)  # pause for 1 second
        number = roll_dice()
        print(f"You got: {number} 🎯")
    elif choice == 'q':
        print("Thanks for playing! 👏")
        break
    else:
        print("Invalid choice! Try again.")


# Main program
play_game()
```

📧 **Project: Email Validator**

🎯 **Goal:**

Check if an entered email address is **valid or invalid** using **functions**.
(This is common in registration forms and login systems.)

💻 **Code Example:**

```
# Email Validator Program


# Function to validate email
```

```
def validate_email(email):
    if "@" not in email or "." not in email:
        return False
    if email.count("@") != 1:
        return False
    if email.startswith("@") or email.endswith("@"):
        return False
    if email.endswith("."):
        return False
    return True


# Function to show result
def show_result(email):
    if validate_email(email):
        print(f"✅ '{email}' is a valid email address.")
    else:
        print(f"❌ '{email}' is not a valid email address.")


# Main program
print("------ Email Validator ------")
user_email = input("Enter your email: ")
show_result(user_email)
```

🧍 **Project: BMI (Body Mass Index) Calculator**

🎯 **Goal:**

Calculate a person's BMI and tell whether they are **Underweight**, **Normal**, **Overweight**, or **Obese** using **functions**.

💻 **Code Example:**

```
# BMI (Body Mass Index) Calculator


# Function to calculate BMI

def calculate_bmi(weight, height):

    bmi = weight / (height ** 2)

    return bmi


# Function to determine BMI category

def bmi_category(bmi):

    if bmi < 18.5:

        return "Underweight"

    elif bmi < 25:

        return "Normal weight"

    elif bmi < 30:

        return "Overweight"

    else:

        return "Obese"


# Function to display result
```

```
def show_bmi_result(name, weight, height):
    bmi = calculate_bmi(weight, height)
    category = bmi_category(bmi)
    print("\n------ BMI REPORT ------")
    print(f"Name: {name}")
    print(f"Weight: {weight} kg")
    print(f"Height: {height} m")
    print(f"BMI: {bmi:.2f}")
    print(f"Category: {category}")
    print("------------------------")

# Main program
name = input("Enter your name: ")
weight = float(input("Enter your weight (kg): "))
height = float(input("Enter your height (m): "))

show_bmi_result(name, weight, height)
```

🏗️ **Project: Currency Converter**

🎯 **Goal:**

Convert money from Indian Rupees (INR) to other currencies using **functions**.

💻 **Code Example:**

```
# Currency Converter


# Function to convert INR to other currencies
def convert_currency(amount, currency):
    if currency == "USD":
        return amount / 83.2      # Example rate: 1 USD = 83.2 INR
    elif currency == "EUR":
        return amount / 90.5      # Example rate: 1 EUR = 90.5 INR
    elif currency == "GBP":
        return amount / 105.3     # Example rate: 1 GBP = 105.3 INR
    elif currency == "JPY":
        return amount * 1.79      # Example rate: 1 INR = 1.79 JPY
    else:
        return None


# Function to display result
def show_result(amount, currency):
    converted = convert_currency(amount, currency)
    if converted is not None:
        print(f"₹{amount} INR = {converted:.2f} {currency}")
    else:
        print("❌ Unsupported currency!")


# Main program
```

```
print("------ Currency Converter ------")

amount = float(input("Enter amount in INR: "))

print("Available currencies: USD, EUR, GBP, JPY")

currency = input("Enter currency to convert into: ").upper()


show_result(amount, currency)
```

🏫 **Project: Student Management System**

🎯 **Goal:**

Create a simple **Student Management System** that allows:

- Adding new students
- Viewing all student records
- Searching a student by roll number
- Deleting a student record

💻 **Code Example:**

```
# Student Management System


students = []  # list to store student data



# Function to add a new student
```

```
def add_student():
    roll = input("Enter Roll Number: ")
    name = input("Enter Name: ")
    course = input("Enter Course: ")
    marks = float(input("Enter Marks: "))

    student = {
        "Roll No": roll,
        "Name": name,
        "Course": course,
        "Marks": marks
    }

    students.append(student)
    print(f"✅ Student {name} added successfully!")


# Function to view all students
def view_students():
    if len(students) == 0:
        print("No student records found!")
    else:
        print("\n------ STUDENT RECORDS ------")
        for s in students:
            print(f"Roll No: {s['Roll No']}, Name: {s['Name']}, Course: {s['Course']}, Marks: {s['Marks']}")
```

```
        print("-----------------------------")


# Function to search student by roll number
def search_student():
    roll = input("Enter Roll Number to search: ")
    found = False
    for s in students:
        if s["Roll No"] == roll:
            print("\n------ STUDENT FOUND ------")
            print(f"Roll No: {s['Roll No']}")
            print(f"Name: {s['Name']}")
            print(f"Course: {s['Course']}")
            print(f"Marks: {s['Marks']}")
            print("---------------------------")
            found = True
            break
    if not found:
        print("❌ Student not found!")


# Function to delete student record
def delete_student():
    roll = input("Enter Roll Number to delete: ")
    for s in students:
```

```
        if s["Roll No"] == roll:
            students.remove(s)
            print(f"🗑️ Student with Roll No {roll} deleted successfully!")
            return
    print("❌ Student not found!")



# Main program menu
def main_menu():
    while True:
        print("\n===== STUDENT MANAGEMENT SYSTEM =====")
        print("1. Add Student")
        print("2. View All Students")
        print("3. Search Student")
        print("4. Delete Student")
        print("5. Exit")

        choice = input("Enter your choice (1-5): ")

        if choice == "1":
            add_student()
        elif choice == "2":
            view_students()
        elif choice == "3":
            search_student()
```

```
    elif choice == "4":
        delete_student()
    elif choice == "5":
        print("Thank you for using Student Management System! 👏")
        break
    else:
        print("Invalid choice! Try again.")



# Start program
main_menu()
```